Introduction and Math Review Practice 1

Module 1 - Introduction & Math Review · Daily practice sheet · No date

Name _____ Date _____

1a In an experiment to study the chemical decomposition, a drug solution was prepared and a sample was obtained at different time points. The drug concentrations in the samples were as follows.

Time (hr)	Concentration (mg/L)
2	294.3
6	208.1
12	123.8
24	43.8
36	15.5
48	5.5

What is the rate constant for the decrease in concentration?

1b A drug solution decomposes by a first-order process with a rate constant of 0.0866 hr^{-1}. The concentration measured 12 hours after the solution was prepared was 123.8 mg/L. What was the initial concentration of the solution?

1c A drug solution decomposes by a first-order process with a rate constant of 0.0866 hr^{-1}. How much time is required for exactly half the solution to decompose?

1d A drug solution decomposes by a first-order process with a rate constant of 0.0866 hr^{-1}. How much time is required for the original solution to decompose by 90 per cent?

2a In an experiment to study the chemical decomposition, a drug solution was prepared and a sample was obtained at different time points. The drug concentrations in the samples were as follows.

Time (hr)	Concentration (mg/mL)
2	338
6	314
12	278
24	206
36	134
48	62

What is the rate constant for the decrease in concentration?

2b A drug solution decomposes by a zero-order process at 6 (mg/mL)/hr, and the sample taken at 2 hours measured 338 mg/mL. What was the initial starting concentration of the solution?

2c A drug solution decomposes by a zero-order process at 6 (mg/mL)/hr from an initial concentration of 350 mg/mL. How much time is required for exactly half the solution to decompose?

2 (final part, unnumbered)

A drug solution with an initial concentration of 350 mg/mL decomposes by a zero-order process at 6 (mg/mL)/hr. How much time is required for the original solution to decompose by 90%?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

1a

Her key

Answer: 0.0866 hr^{-1}

WHAT THE STEM GIVES

C_1	208.1 mg/L	the earlier concentration, at t_1 = 6 hr, in the numerator
t_1	6 hr	the earlier time
C_2	43.8 mg/L	the later concentration, at t_2 = 24 hr
t_2	24 hr	the later time; $t_2 - t_1$ = 18 hr in the denominator
other rows	294.3, 123.8, 15.5, 5.5 mg/L	*not needed for k: they confirm first order, equal ratios over equal spans*

THE WORKING, ONE LINE AT A TIME

SETUP

Ratios over equal intervals are constant: $\frac{123.8}{43.8}$ = 2.83 across the 12 hours from 12 to 24 hr, and $\frac{43.8}{15.5}$ = 2.83 across the 12 hours from 24 to 36 hr, so a fixed fraction is lost per unit time and the process is first order.

The order must be settled first, because the two orders give rate constants with different units and equations. Ratios must be compared over equal spans (a 6-hour and a 12-hour ratio are not expected to match), and two consecutive 12-hour spans both giving 2.83 is the first-order condition.

SETUP

For a first-order process, $\ln C = \ln C_0 - kt$, so $k = \frac{\ln(C_1 \div C_2)}{t_2 - t_1}$.

Writing the integrated first-order equation for two measured points and subtracting cancels $\ln C_0$, so the unknown starting concentration drops out and k depends only on values in the table. Dr. Mosley writes this difference of logarithms as the logarithm of the ratio.

ALGEBRA

Using t_1 = 6 hr with C_1 = 208.1 mg/L and t_2 = 24 hr with C_2 = 43.8 mg/L: Δt = 18 hr, and $k = \frac{\ln(208.1 \div 43.8)}{18}$ hr = $\frac{1.55838}{18}$ hr.

These are the two points in her own key, and the 18-hour separation spans most of the decline, which reduces the effect of measurement scatter. The concentration units cancel inside the logarithm, leaving a pure number divided by hours: reciprocal hours.

ROUND

k = 0.08658 hr^{-1}, reported as 0.0866 hr^{-1}.

Dividing a dimensionless logarithm by a time gives reciprocal time, the unit of a first-order rate constant. Dr. Mosley reports rate constants to four decimal places (her key prints 0.0866 per hour), and a leading decimal with no digit before it loses half the marks on her paper.

DOES THE NUMBER MAKE SENSE?

Each 12-hour span cuts the concentration to about a third (123.8 → 43.8 → 15.5), so 12 hr is between one and two half-lives and t½ is near 8 hr; k has to come out near 0.0866 hr^{-1}, a small positive number.

1b Her key

Answer: 350 mg/L

WHAT THE STEM GIVES

k	0.0866 hr^{-1}	the exponent kt = 0.0866 × 12 = 1.0392
t	12 hours	the time in the exponent
C	123.8 mg/L	the later concentration, multiplied by e^{kt} to go back to C_0

THE WORKING, ONE LINE AT A TIME

SETUP

$C = C_0\, e^{-kt}$, so $C_0 = C\, e^{+kt}$.

The integrated equation runs forward from the start, but the unknown is at the start, so it is rearranged to move backwards: multiplying both sides by e^{+kt} isolates C_0. Keeping the minus sign returns a number smaller than the measured concentration, which cannot be a starting value.

ALGEBRA

$kt = 0.0866\ hr^{-1} \times 12\ hr = 1.0392$, which is dimensionless.

Reciprocal hours multiplied by hours cancel, which they must, because an exponent has to be a pure number. Checking that cancellation is the quickest guard against having mixed a rate constant in reciprocal hours with a time in days or minutes. The product itself has no units.

ALGEBRA

$C_0 = 123.8\ mg/L \times e^{1.0392} = 349.98\ mg/L$.

The exponential factor $e^{1.0392}$ is 2.8270, which is greater than one because the exponent is positive, so the answer is larger than the measured concentration, as it must be for a concentration that has been falling for 12 hours. Milligrams per litre multiplied by a pure number stay milligrams per litre.

ROUND

$C_0 = 350\ mg/L$.

Dr. Mosley reports this value as 350 mg/L, to three significant figures, which matches the precision of the tabulated concentrations. The computed 349.98 rounds to that value without any further adjustment.

DOES THE NUMBER MAKE SENSE?

With k = 0.0866 hr^{-1} the half-life is about 8 hr, so 12 hours is more than one half-life: C_0 must be more than double 123.8 mg/L and less than four times it, and 350 mg/L fits.

1c Her key

Answer: 8 hr

WHAT THE STEM GIVES

k	0.0866 hr^{-1}	the denominator of $\frac{0.693}{k}$

THE WORKING, ONE LINE AT A TIME

SETUP

$t_{½} = \frac{0.693}{k}$ for a first-order process.

Setting the concentration to half its starting value in $\ln C = \ln C_0 - kt$ leaves $\ln 2$, which is 0.693, divided by the rate constant. The starting concentration cancels, so a first-order half-life does not depend on it. This relation is not printed on the exam equation sheet.

ALGEBRA

$t_{½} = \frac{0.693}{0.0866}\ hr^{-1} = 8.002$ hr.

Dividing a pure number by a quantity in reciprocal hours gives hours, which is the unit a half-life must have. The rate constant sits in the denominator because a faster process has a shorter half-life, so the two quantities move in opposite directions.

ROUND

$t_{½} = 8$ hr.

Dr. Mosley reports this as 8 hours, to the whole hour. The answer is a time and carries no reciprocal unit; hours to the minus one belongs to the rate constant that was given, not to the half-life that was asked for.

> **DOES THE NUMBER MAKE SENSE?**
> The half-life is 0.693 over k. A k near one tenth per hour gives a half-life near 7 hr; 8 hr fits, and k × $t_{½}$ must return 0.693.

1d Her key

Answer: 26.6 hr

WHAT THE STEM GIVES

k	0.0866 hr^{-1}	the denominator of $\frac{\ln(10)}{k}$
fraction left	90 per cent	decomposed, so 10 per cent remains and ln(10) sits in the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

Decomposing by 90 per cent leaves 10 per cent, so $C = 0.1\ C_0$.

The stem states how much has gone and the equation is written in terms of how much remains, so the percentage has to be turned round before it is used. Reading 90 per cent straight into the equation as the remaining fraction gives $\frac{\ln(1 \div 0.9)}{0.0866}$ $hr^{-1} = 1.2$ hours instead of 26.6.

SETUP

$0.1\ C_0 = C_0\ e^{-kt}$, so $t = \frac{\ln(1 \div 0.1)}{k}$.

The starting concentration cancels from both sides, which is why no concentration value is needed to answer a percentage question for a first-order process. Rearranging for time puts the logarithm of the reciprocal fraction over the rate constant.

ALGEBRA

$t = \frac{\ln(10)}{0.0866}\ hr^{-1} = \frac{2.302585}{0.0866}\ hr^{-1}$.

The logarithm of ten is a pure number, and dividing it by a quantity in reciprocal hours leaves hours. The fraction remaining is not a clean power of one half here, so half-life counting will not reach the answer exactly and the logarithmic route is needed.

ROUND

$t = 26.589$ hr, reported as 26.6 hr.

Dr. Mosley reports this to one decimal place, as 26.6 hours. As a check, 90 per cent decomposition is a little more than three half-lives, which for an 8-hour half-life would be 24 hours, and the answer sits just above that.

> **DOES THE NUMBER MAKE SENSE?**
> Ten per cent remaining lies between one eighth and one sixteenth, so the time is between three and four half-lives of 8 hr, a little over 24 hr; 26.6 hr fits.

2a Her key

Answer: 6 (mg/mL)/hr

WHAT THE STEM GIVES

C_1	338 mg/mL	the earlier concentration, at $t_1 = 2$ hr
t_1	2 hr	the earlier time
C_2	206 mg/mL	the later concentration, at $t_2 = 24$ hr
t_2	24 hr	the later time; $t_2 - t_1 = 22$ hr in the denominator
other rows	314, 278, 134, 62 mg/mL	*not needed for k: they confirm zero order, equal drops over equal spans*

THE WORKING, ONE LINE AT A TIME

SETUP

Successive losses are constant: 338 − 314 = 24 mg/mL over 4 hr, and 314 − 278 = 36 mg/mL over 6 hr, both 6 mg/mL per hour, so the process is zero order.

The differences between concentrations, taken over the intervals that separate them, are what identify a zero-order process. Here the same quantity is lost in every hour regardless of how much remains. Testing ratios instead would give 1.08 and then 1.13, which are not equal, confirming that the process is not first order.

SETUP

For a zero-order process, $C = C_0 - kt$, so $k = -\dfrac{C_2 - C_1}{t_2 - t_1}$.

The integrated zero-order equation is a straight line, so its rate constant is the slope with the sign reversed. The minus sign is there because the concentration falls while k is reported as a positive quantity; a rate constant is never negative, in either order.

ALGEBRA

Using $t_1 = 2$ hr with $C_1 = 338$ mg/mL and $t_2 = 24$ hr with $C_2 = 206$ mg/mL: $k = -(206 - 338)$ mg/mL ÷ $(24 - 2)$ hr = 132 mg/mL ÷ 22 hr.

These are the two points Dr. Mosley takes in her key. Milligrams per millilitre divided by hours gives milligrams per millilitre per hour, which is concentration per unit time, the unit a zero-order rate constant must carry.

ROUND

k = 6 (mg/mL)/hr.

The division is exact at 6, and her key prints 6 mg/mL per hour with no further rounding. Because the process is zero order, this number is the quantity lost every hour, unchanged from the first hour to the last.

> **DOES THE NUMBER MAKE SENSE?**
> From 2 to 24 hr the concentration drops 338 → 206 mg/mL, 132 mg/mL in 22 hr, so the rate is 132 over 22, 6 (mg/mL)/hr: a fixed drop per hour, not a fixed fraction.

2b Her key

Answer: 350 mg/mL

WHAT THE STEM GIVES

k	6 (mg/mL)/hr	the kt added back: 6 × 2 = 12 mg/mL
t	2 hours	the sample time, multiplied by k
C	338 mg/mL	the 2-hour concentration that 12 mg/mL is added to

THE WORKING, ONE LINE AT A TIME

SETUP

$C = C_0 - kt$, so $C_0 = C + kt$

The zero-order integrated equation is a straight line, and the initial concentration is its intercept. Rearranging to put C_0 on its own adds back what was lost, so the sign in front of the rate constant flips from minus to plus.

ALGEBRA

$C_0 = 338 \text{ mg/mL} + (6 \text{ (mg/mL)/hr})(2 \text{ hr}) = 338 + 12 = 350 \text{ mg/mL}$

Milligrams per millilitre per hour multiplied by hours leaves milligrams per millilitre, so the two terms can be added. Six milligrams per millilitre disappear in each of the first two hours, which is the 12 added back.

ROUND

$C_0 = 350 \text{ mg/mL}$

Her key prints 350 mg/mL. The value is exact, so no rounding is involved, and it is the number the half-life part of this problem then uses.

DOES THE NUMBER MAKE SENSE?

Two hours at 6 mg/mL per hour is 12 mg/mL, so C_0 is 12 above 338, 350 mg/mL; the start must exceed the 2-hour sample.

2c Her key

Answer: 29.2 hr

WHAT THE STEM GIVES

k	6 (mg/mL)/hr	the 2k in the denominator: 2 × 6 = 12
C_0	350 mg/mL	the numerator of $\frac{C_0}{2k}$

THE WORKING, ONE LINE AT A TIME

SETUP

$t½ = \frac{C_0}{2k}$ for a zero-order process.

Setting the concentration to half its starting value in $C = C_0 - kt$ gives $\frac{C_0}{2} = kt$, and dividing by k isolates the time. C_0 does not cancel, so a zero-order half-life depends on where the process started and is not a fixed property of the drug.

ALGEBRA

$t½ = 350 \text{ mg/mL} \div [2 \times 6 \text{ (mg/mL)/hr}] = \frac{350}{12} \text{ hr.}$

Milligrams per millilitre divided by milligrams per millilitre per hour leaves hours, since the concentration units cancel and the reciprocal hour inverts. The factor of two appears because only half of the starting concentration has to be removed.

ROUND

$t½ = 29.167$ hr, reported as 29.2 hr.

Dr. Mosley reports this to one decimal place, as 29.2 hours. As a check, the table for this solution reads 206 mg/mL at 24 hours and 134 mg/mL at 36 hours, so the concentration passes 175 mg/mL between those two times, which brackets the answer.

DOES THE NUMBER MAKE SENSE?

> Half of 350 mg/mL is lost at 6 mg/mL each hour, so t½ is 350 over 12, about 29.2 hr; it is half the time to lose everything, so it is well under 60 hr.

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2 (final part, unnumbered) Her key

Answer: 52.5 hr

WHAT THE STEM GIVES

C_0	350 mg/mL	the start; 90% gone leaves C = 35 mg/mL, so $C_0 - C = 315$
k	6 (mg/mL)/hr	the denominator of $\frac{315}{6}$
fraction	90%	decomposed, which fixes C = 35 mg/mL

THE WORKING, ONE LINE AT A TIME

SETUP

90 per cent decomposed leaves 10 per cent, so C = 0.10 × 350 mg/mL = 35 mg/mL

The question names what has gone and the equation is written in terms of what is left, so the percentage has to be turned round before anything is substituted. A tenth of 350 is 35, and that is the concentration the solution has to fall to.

SETUP

$$C = C_0 - kt, \text{ so } t = \frac{C_0 - C}{k}$$

Rearranging the zero-order line puts the time on its own. The numerator is the amount of concentration that has to disappear and the denominator is how much disappears each hour, so the quotient is a number of hours.

ALGEBRA

$$t = (350 - 35) \text{ mg/mL} \div (6 \text{ (mg/mL)/hr}) = \frac{315}{6} = 52.5 \text{ hr}$$

Milligrams per millilitre divided by milligrams per millilitre per hour leaves hours. The 315 is what must be lost, and at 6 per hour that takes 52.5 hours.

ROUND

t = 52.5 hr

Her key prints 52.5 hr, to one decimal place. As a check, the table for this solution reads 62 mg/mL at 48 hours, so 35 mg/mL is reached a few hours after the last sample, which brackets the answer.

> **DOES THE NUMBER MAKE SENSE?**
> Ninety per cent of 350 mg/mL is 315 mg/mL, and at 6 mg/mL per hour that takes 315 over 6, 52.5 hr; it must be less than the time to lose all 350, under 60 hr.
